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Report for Computer Vision in Archaeology Training School

From 14 to 18 September 2024, I had the opportunity to participate in the *Computer Vision in Archaeology* Training School, hosted by the Archaeological Information System of the Czech Republic (AIS CR) at the Institute of Archaeology of the Czech Academy of Sciences in Brno. The event was co-organised under the framework of the EU-funded *Advancing Frontier Research in the Arts and Humanities* (ATRIUM) Project and the *Managing Artificial Intelligence in Archaeology* (MAIA) COST Action [CA23141].

This intensive five-day programme aimed to equip participants with skills ranging from foundational principles to advanced applications of computer vision within archaeological research. The opening day was dedicated to an overview of computer vision, the specific research questions it can address in archaeology, the integration of AI coding assistants, and core Python programming. Practical exercises conducted via Google Colab allowed me to familiarise myself directly with these computational tools. The subsequent four days focused on key domains of application, including the analysis of metal artefacts and coinage, photographs of geoarchaeological thin-sections, 3D and digital models of lithics, satellite remote sensing, and rock art documentation. Following in-depth theoretical lectures, we gained extensive hands-on experience in image pre-processing, dataset annotation for training, automated segmentation, and the deployment of diverse pattern-recognition algorithms. These academic sessions were complemented by social activities that fostered valuable networking opportunities with both instructors and fellow researchers. Furthermore, several participants had the chance to present ongoing research, receiving constructive and insightful feedback from the group.

Overall, the ATRIUM-MAIA Training School was exceptionally beneficial to my professional development. It significantly expanded my understanding of computer vision and kept me abreast of the rapid methodological advances made in recent years, particularly following the emergence of Large Language Models (LLMs). This training will undoubtedly have a profound impact on my upcoming research on the automated detection of cropmarks and archaeological features in remote sensing data, a project for which I am currently seeking funding through an MSCA Postdoctoral Fellowship.

Regarding potential enhancements to an otherwise outstanding programme, incorporating a more comprehensive overview of the core literature directly contextualising the case studies would have provided even greater depth. Nevertheless, the dedicated Zotero library maintained

by the MAIA Action serves as an excellent resource to bridge this gap. Additionally, circulating preparatory materials prior to the school, particularly on basic programming concepts in Python, could help level the playing field and address prerequisite knowledge gaps among participants.

To conclude, the Training School proved to be an immensely stimulating and rewarding experience, markedly strengthening my capabilities in applying automated vision methods to archaeological research. The practical competencies and theoretical insights acquired will be pivotal in driving my future research forward, ultimately facilitating a deeper understanding of past human activities through advanced digital methodologies.